BOWEN YANG

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EDUCATION

Shanghai Jiao Tong University (SJTU)
B. Eng in Information Engineering, B.A in French Studies Graduating Expected June 2027
SJTU Paris Elite Institute of Technology
GPA: **91.7/100**; Rank: **7/110**
Advisor: **Prof. Junchi Yan**, **Prof. Xiaosong Jia**

RESEARCH EXPERIENCE

GuidedVLA: Specifying Task-Relevant Factors via Plug-and-Play Action Attention Specialization
Robotics: Science and Systems (RSS) 2026 Project Page | arXiv
Research Assistant, RethinkLab, SJTU Sep 2025 – Jan 2026

- Led the **codebase implementation and simulation experiments** for GuidedVLA.
- Implemented **specialized action-attention heads** for object grounding, temporal skill recognition, and depth-based geometric perception in the π_0 action decoder.
- Integrated a **ControlNet-style residual adapter** for plug-and-play factor supervision while preserving the pretrained VLA policy pathway, enabling stable guidance injection without disrupting the base policy.
- Boosted average success rates from **68.2% to 75.4%** on **LIBERO-Plus** and **77.4% to 90.6%** on **RoboTwin 2.0**, while improving training throughput by **30%+** via mixed precision and **torch.compile**.

ROBOTICS COMPETITIONS & PROJECTS

RoboMaster University Championship
Vice-Captain & Algorithm Lead, Team Jiao Loong Oct 2023 – Aug 2025

- As **Vice-Captain and Algorithm Lead**, managed the R&D of all algorithm projects across SLAM, navigation, perception for Team Jiao Loong, and won **National Champions in 2024 & 2025**.
- Led development of a **LiDAR–inertial SLAM pipeline** (Fast-LIO2 based) with GICP map registration and fully PTP-synchronized LiDAR timing to improve robustness under fast robot motion.
- Built a **motion planning system** combining **kinodynamic A* and MPC** and achieve low-latency tracking on embedded platforms under real-world competition constraints.
- Deployed **TensorRT-accelerated YOLO** on **Jetson Orin** and integrated an **EKF-based motion prediction** module for reliable enemy-state estimation under occlusion and high-speed dynamics.

AWARDS AND HONORS

- **National Scholarship** 2024, 2025
  Ministry of Education of China; top 0.2% of students nationwide.
- **National Champion**, RoboMaster University Championship 2024, 2025
  Core contributor to SLAM, navigation, perception, and embedded systems.
- **RM Award (Algorithm)**, RoboMaster University Championship 2025
  Individual award for SLAM and navigation algorithm design of the Sentry robot.
- **Finalist (Top 2%)**, Mathematical Contest in Modeling (MCM) 2025
  Problem B (Discrete). Environmental Dynamic System Evolution and Multi-Objective Optimization Modeling
- **NOI Winter Camp Bronze**; **USACO Platinum Division** 2020–2021
  Advanced training in algorithms and competitive programming.

TECHNICAL SKILLS

Programming: Python, C, C++
Frameworks & Tools: PyTorch, TensorFlow, ROS/ROS2, OpenCV, TensorRT, Docker, Git, LaTeX
Languages: English (TOEFL R6/L5.5/S6/W5), French (DELF B2), Cantonese (Conversational)